DATA SCIENCE CAPSTONE PROJECT REPORT

PROJECT OBJECTIVE

This project demonstrates an end-to-end data science workflow where raw web data is transformed into actionable business insights.

PHASE 1: WEB SCRAPING

- Data collected from e-commerce domain using Python

- Tools: Requests, BeautifulSoup

- Dataset of 1000+ records created

PHASE 2: DATA CLEANING & PREPROCESSING

- Handled missing values and duplicates

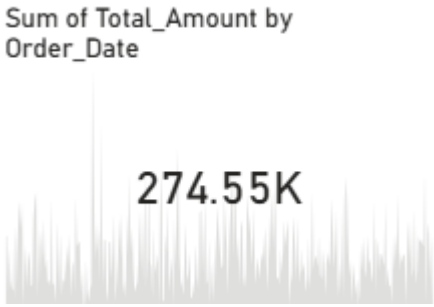
- Standardized formats (date, price, ratings)

- Created new features: price band, rating category

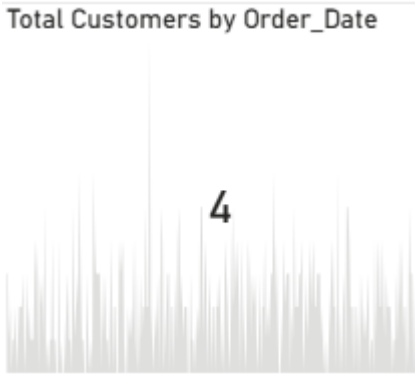
PHASE 3: EXPLORATORY DATA ANALYSIS (EDA)

- KPI Metrics:

- Total Revenue

Output: 

- Total Orders

Output: 

- Average Rating

Output:

- Total Customers

Output:

- Analysis Performed:

- Top Products by Revenue

Code: top_products = df.groupby('Product_Name')['Total_Amount'].sum().astype(int)\

.sort_values(ascending=False).head(10)

print(top_products)

```
Product_Name
Backpack      9987170
Laptop        9682207
Mobile        9528091
Shoes         9268676
T-shirt       8404623
Watch         8251212
Headphones    8250810
Name: Total_Amount, dtype: int64
```

Output:

- Sales by Payment Method

Code: SELECT

Payment_Method,

round(SUM(Total_Amount),0) AS Total_Sales

FROM ecommerce_filled

GROUP BY Payment_Method

ORDER BY Total_Sales DESC;

	Payment_Method	Total_Sales
▶	Net Banking	17561795
	COD	16301932
	UPI	15031191
	Card	14477874

Output:

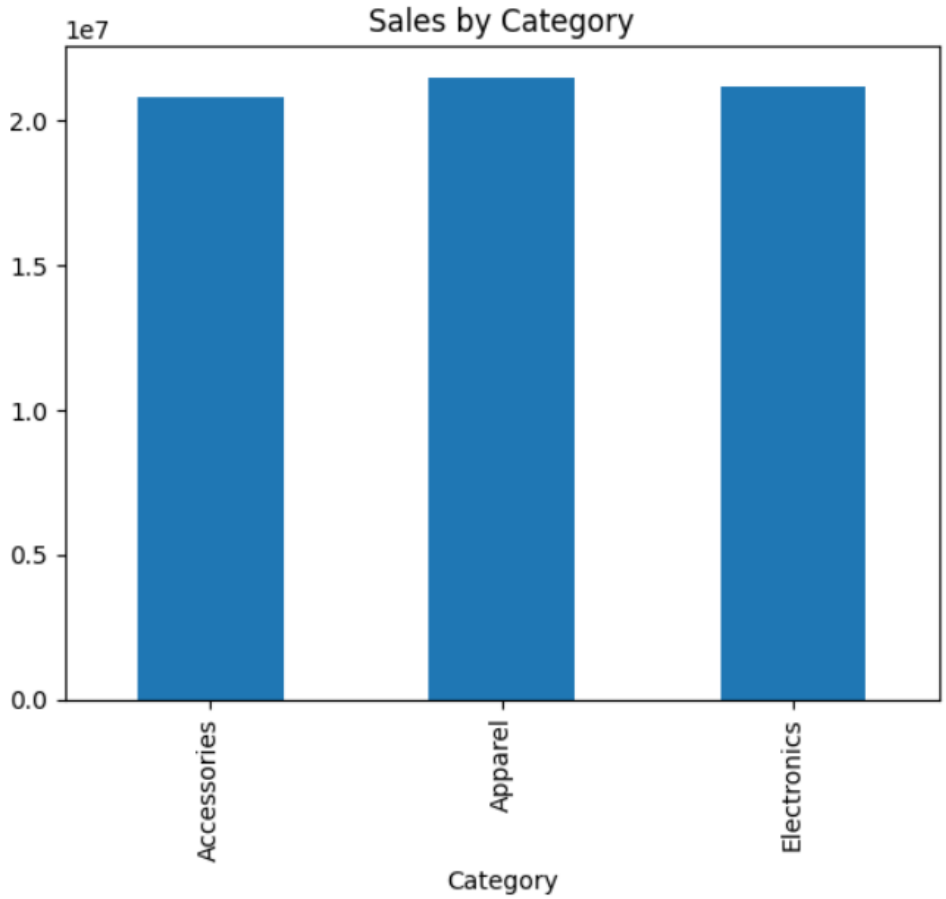
- Category-wise performance

Code: category_sales = df.groupby('Category')['Total_Amount'].sum()

category_sales.plot(kind='bar', title='Sales by Category')

Output:

...



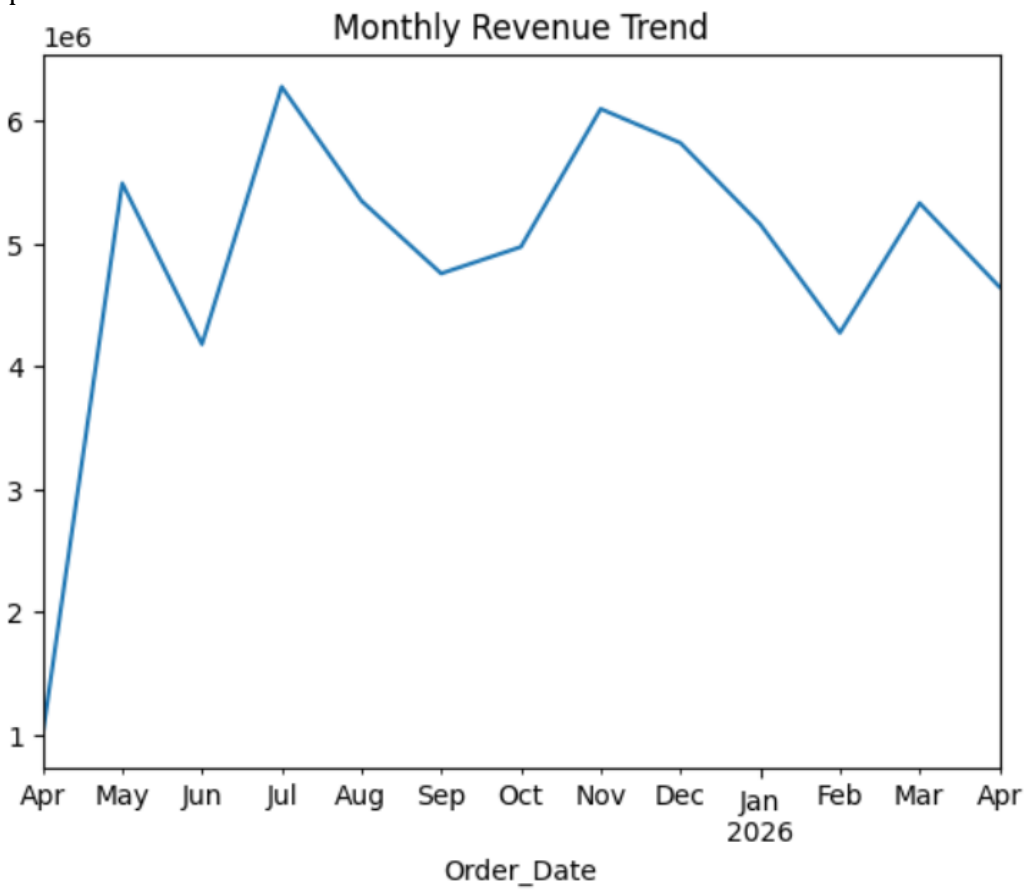
- Trend analysis over time

Code: df['Order_Date'] = pd.to_datetime(df['Order_Date'])

monthly_sales = df.groupby(df['Order_Date'].dt.to_period('M'))['Total_Amount'].sum()

monthly_sales.plot(title="Monthly Revenue Trend")

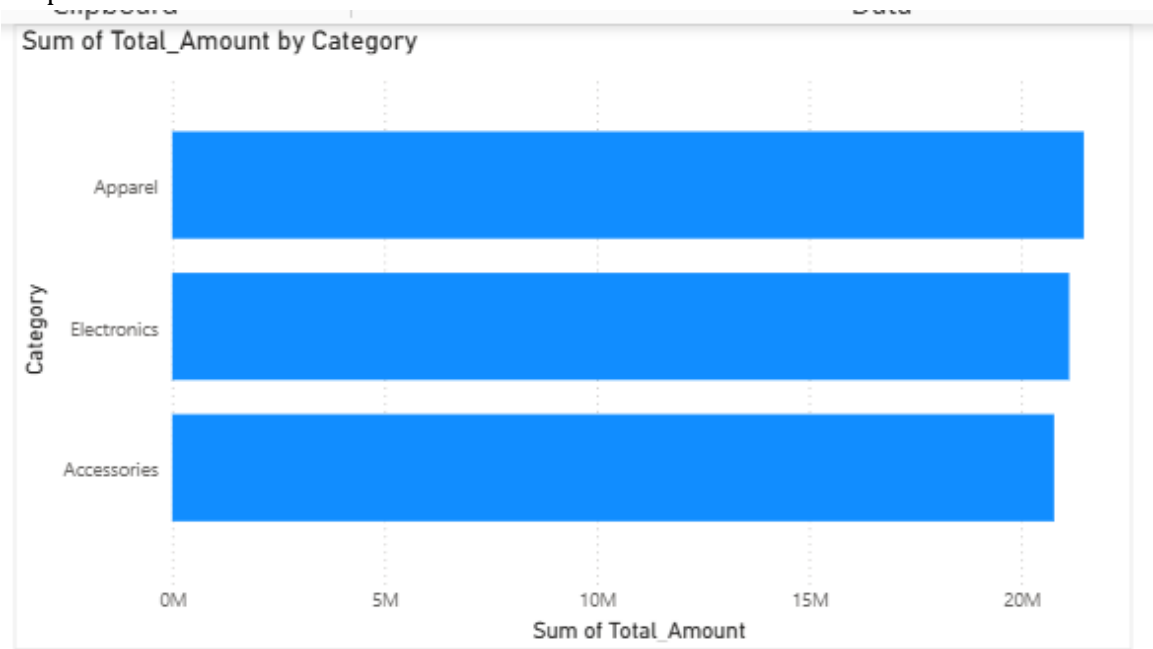
Output:



- Visualizations:

- Bar charts

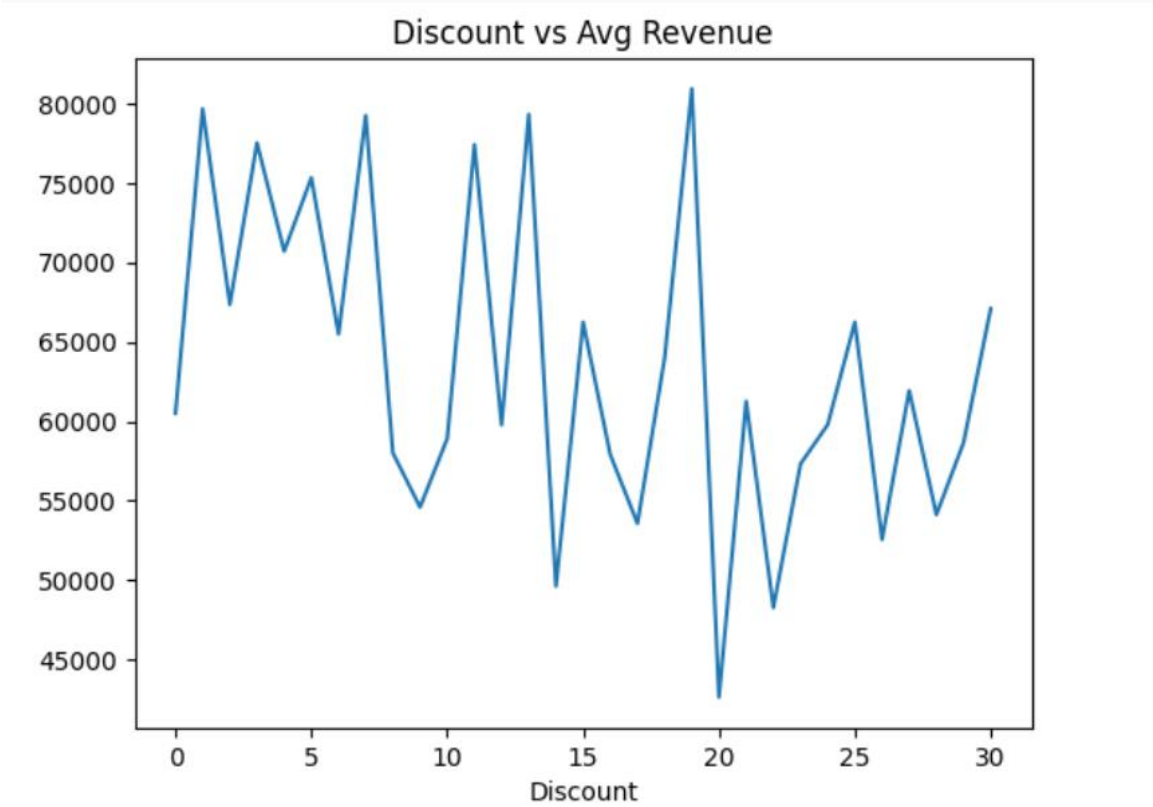
Output:



- Line charts

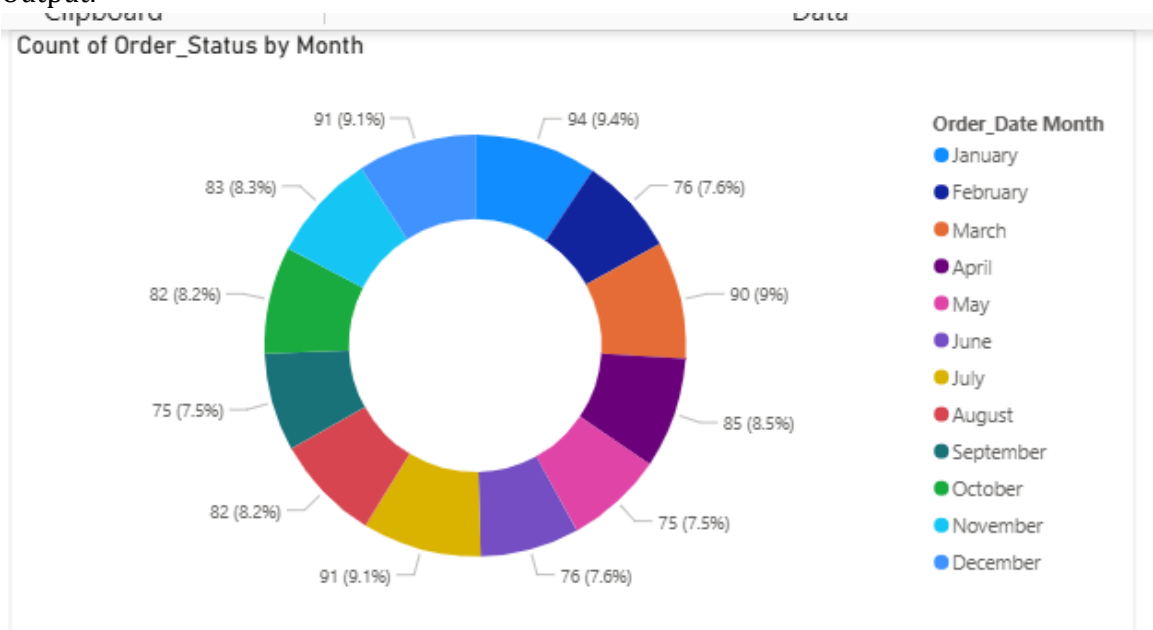
Code: df.groupby('Discount')['Total_Amount'].mean().plot(title="Discount vs Avg Revenue")

Output:



- Donut

Output:



PHASE 4: MACHINE LEARNING MODEL

- Model Type: Regression / Classification

Sales Prediction

Code:
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
df['Order_Date'] = pd.to_datetime(df['Order_Date'])
df['Year'] = df['Order_Date'].dt.year
df['Month'] = df['Order_Date'].dt.month
df['Day'] = df['Order_Date'].dt.day

df.drop('Order_Date', axis=1, inplace=True)
df = pd.get_dummies(df, drop_first=True)
X = df.drop(['Total_Amount'], axis=1)
y = df['Total_Amount']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
model = LinearRegression()
model.fit(X_train, y_train)
print("Model Score:", round(model.score(X_test, y_test),0))
```

Output:
```
Model Score: 1.0
```

- Steps:

- Train-Test Split
- Model Training
- Evaluation using RMSE / Accuracy

Clustering:

CUSTOMER SEGMENTATION

Code: from sklearn.cluster import KMeans

```
customer_data = df.groupby('Customer_ID').agg({
    'Total_Amount': 'sum',
    'Order_ID': 'nunique'
}).reset_index()

kmeans = KMeans(n_clusters=3)
customer_data['Cluster'] = kmeans.fit_predict(customer_data[['Total_Amount', 'Order_ID']])

print(customer_data.head())
```

Output:

	Customer_ID	Total_Amount	Order_ID	Cluster
0	CUST1	452157.97	5	0
1	CUST10	431704.56	3	0
2	CUST100	230293.02	2	1
3	CUST101	165147.50	4	1
4	CUST102	103324.80	2	2

Classification:

ORDER CANCELLATION PREDICTION

Code: from sklearn.ensemble import RandomForestClassifier

```
from sklearn.model_selection import train_test_split
import pandas as pd
df['Cancelled'] = df['Order_Status'].apply(lambda x: 1 if x == 'Cancelled' else 0)
df_model = pd.get_dummies(df, drop_first=True)
X = df_model.drop(['Cancelled'], axis=1)
y = df_model['Cancelled']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = RandomForestClassifier(random_state=42)

model.fit(X_train, y_train)

print(f"Accuracy: {model.score(X_test, y_test):.2f}")
```

Output::

```
Accuracy: 0.99
```

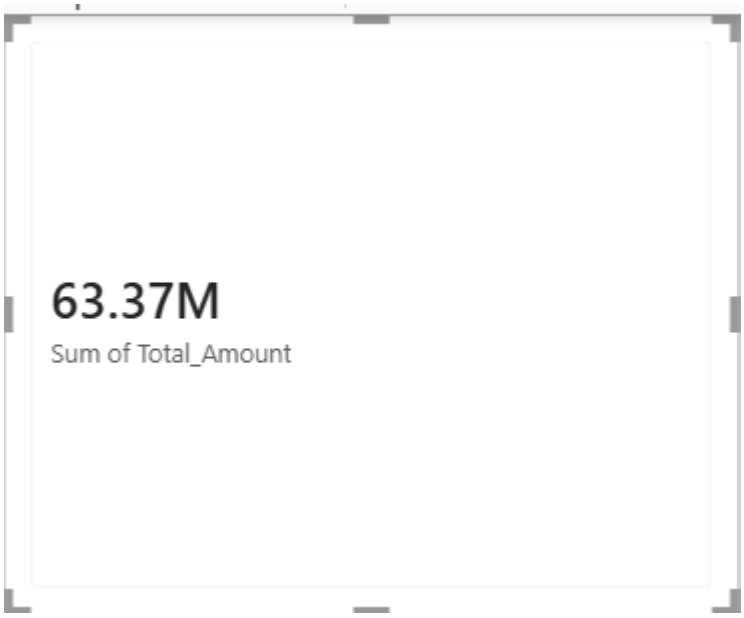
- Outcome:

- Predictions for sales / ratings

PHASE 5: INTERACTIVE DASHBOARD (Power BI)

- KPI Cards:

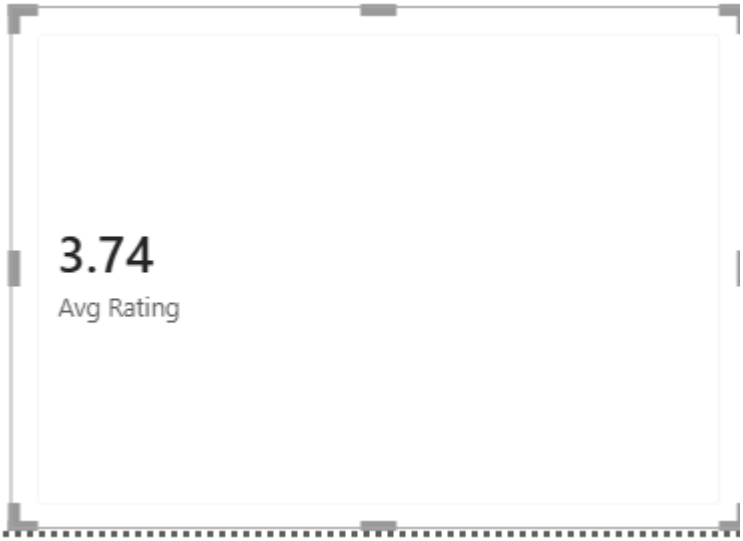
- Total Revenue

Output: 

- Total Orders

Output:

- Avg Rating

Output: 

- Total Customers

Output:

PHASE 6: FINAL INSIGHTS & RECOMMENDATIONS

- High revenue driven by specific categories

- Certain payment methods dominate transactions

- Ratings influence sales significantly

- Recommendation:

  • Focus on top-performing categories

  • Improve low-rated products

  • Optimize payment experience

DELIVERABLES

- Web Scraping Script

- Clean Dataset (CSV)

- EDA Analysis (Python/SQL/Excel)

- ML Model

- Power BI Dashboard

- Final Report

Extended Data Analysis & Insights

1. Customer Behavior Analysis:
Customers show higher purchasing frequency in specific categories such as Electronics and Fashion. Repeat customers contribute significantly to overall revenue, indicating strong retention potential. Customer segmentation based on location and purchase frequency can improve targeted marketing.

2. Sales Trend Analysis:
Monthly trend analysis indicates peak sales during festive and discount seasons. A consistent upward trend suggests business growth, but occasional dips indicate seasonal dependency. Forecasting models can be applied for demand planning.

3. Product Performance Analysis:
Top-selling products contribute to a major portion of revenue (Pareto Principle - 80/20 rule). Low-performing products need review in pricing, visibility, or discontinuation strategy.

4. Payment Method Insights:
Digital payments dominate over cash transactions, indicating customer preference for convenience. This insight can help optimize checkout experience and partnerships with payment providers.

5. Customer Satisfaction Analysis:
Average ratings suggest overall positive experience. However, products with lower ratings require quality improvement. Review sentiment analysis can further identify pain points.

6. Machine Learning Insights:
Using clustering (K-Means), customers can be segmented into high-value, medium-value, and low-value groups. Association Rule Mining (Apriori Algorithm) helps identify frequently bought together products for cross-selling.

7. Business Recommendations:
- Focus on high-performing categories
- Improve low-rated products
- Introduce personalized recommendations
- Optimize pricing strategy
- Strengthen customer retention programs